Mess Waste Advisor

Measuring what hostel messes throw away, and turning it into what to cook tomorrow

Latent 48 · Bring the Physical World to AI

Team Liabilities

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[PHOTO: REAL MESS WASTE BIN]

Physical evidence benchmark · Pre-meal audit intake record

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THE PROBLEM

Hostel messes waste tens of kilos of food every meal, and no one measures what, how much, or why

THE PROBLEM

- **Cooking by habit.** Each mess cooks breakfast, lunch and dinner for hundreds of students. Quantities are set by routine, not by records of what was eaten.
- **Waste leaves no trace.** Unserved food and plate scrapings go into bins without being weighed, so the same dishes are over-cooked and rejected week after week.
- **No feedback loop.** Without a record, the mess can't tell whether waste comes from cooking too much or from students not eating a dish, and those need different fixes.

WHO HAS THIS PROBLEM

- **Primary user:** the hostel mess manager, who decides every day how much of each item to cook and what goes on the menu.
- **Decision-makers:** the hostel mess committee, which approves menu changes.
- **What they lack:** a simple, repeatable record of waste per meal, split by type, that they can compare across days.

WHY WE CHOSE IT

- A real physical workflow, repeated 3 times a day in every hostel, and observable without disrupting anyone.
- The hackathon brief names it directly: "helping mess managers reduce … plate waste across hostels."
- Cheap to measure (a scale, a phone camera and a form), and the result leads straight to a decision: cook less, or change the dish.

Our question: For each meal, what is wasted, how much, why, and what should the mess change: the menu or the quantity cooked?

Unserved food = cooked but never served or taken (includes counter leftovers). Plate waste = served, then thrown from plates.

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AUDIT & SOLUTION

We audited post-meal waste across 9 hostel messes with photo evidence, and built a tool that turns each photo into a decision

THE MESS WORKFLOW AND WHERE WE MEASURE

Cook → Serve → ① Unserved food (weighed at close of service) → Eat → Bin → ② Plate waste (weighed from the waste bin)

HOW WE COLLECTED THE DATA

Item	Detail
Hostels	9: Barak, Brahma, Dihing, Disang, Kameng, Kapili, Lohit, Siang, Umiam
Meals	Breakfast, lunch, dinner
Events	24 unique observation events (55 raw rows; repeat photos merged)
Window	[collection dates]
Evidence	A waste-bin photo linked to every event
Measurement	Scale where possible; otherwise volume estimate (2.5 ft vessel ≈ 46 kg; rice waste 1.4 g/mL)
Format	Structured table with schema, exported as Parquet

DATA HONESTY:

- **Observed:** most unserved-food values and about half of plate-waste values (scale readings such as 38.255 kg and 63.46 kg).
- **Estimated:** at least 1 value (Kameng lunch, 46 kg).
- **Synthetic:** none.
- **Missing:** left blank, not filled in (plate waste missing in 12 of 24 events).

WHAT WE BUILT: MESS WASTE ADVISOR

1. **Upload:** after each meal, the manager uploads a waste photo and that day's menu.
2. **Analyze:** estimates waste in kg for each category, identifies menu items in the waste, and compares it with the same meal on earlier days.
3. **Recommend:** reports what was wasted, how much and why, and which menu items to resize or replace and how much to cook that day.
4. **Record:** saves every entry with its photo and exports an open Parquet dataset with a data dictionary.

ROLE OF AI: Image analysis estimates waste and recognises dishes from the day's menu; recommendations follow transparent rules applied to the recorded history.

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FINDINGS & IMPACT

Plate waste averaged 44.8 kg per event against 13.3 kg of unserved food, so what students leave on plates is the bigger lever

24

observation events across 9 hostels

44.8 kg

average plate waste per event (12 events · range 22.4–74.8 kg)

13.3 kg

average unserved food per event (22 events · range 2–35 kg)

100%

of events backed by a photograph

RECORDED WASTE PER HOSTEL (KG)

[Chart: grouped bars per hostel, unserved vs plate waste, values from the final CSV]



Averages are over different events (22 with unserved, 12 with plate waste). Totals per mess, not per student.

WHAT THE DATA TELLS THE MESS MANAGER

- Plate waste is the larger stream, so portion size and dish choice come before cooking volume.
- Waste varies sharply between hostels for the same meal type, so the leanest messes can serve as the benchmark.
- Every number now has a photo behind it, so waste can be checked instead of argued about.

WHAT CHANGES, STARTING TOMORROW

- **Log every meal:** a photo plus a weight takes under two minutes and builds the history.
- **Menu optimisation:** dishes that keep coming back on plates get flagged to resize or replace.
- **Quantity optimisation:** when unserved food repeats on the same day and meal, the tool recommends cooking less of those items.

Limits: about one day of observations; uneven coverage (dinner in 3 of 9 hostels);

no headcount, so no per-student comparison; some weights are volume estimates.
Demo: [link] · **Code and data:** [link]

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